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Compensation of Memory-Effects in PDM-based Transmitters for Digital Audio Broadcast

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***Abstract*— This paper examines existing *amplitude modulation* (AM) broadcast technology based on the *envelope elimination and restoration* (EER) amplification technique to determine its feasibility in accepting the new terrestrial *digital audio broadcast* (DAB) standards. When applying the hybrid AM DAB signals to EER amplifiers based on *pulse duration modulation* (PDM), digital signal processing algorithms are proposed to improve the overall performance of the system to meet the AM spectral mask.**

An investigation of the effects the various components of the EER amplifier have on the DAB signals is conducted to identify the most critical components in the system that contribute to the spectrum regrowth. It is found that the realistic envelope reconstruction filter in the envelope path of the EER amplifier requires specialized signal processing to extend the envelope path bandwidth. In particular, an LMS-based predistorter is proposed to improve filter frequency response and allow the DAB signalling to meet the required spectral mask.

I. Introduction

In the past few decades, communication technology has experienced rapid change as digital systems have replaced existing analog based ones. Notwithstanding, one area that has remained relatively unchanged is *terrestrial audio broadcasting* (TAB). *Amplitude modulation* (AM) and *frequency modulation* (FM) transmission systems today are still using technology that was designed some 40 years ago, but with the recent interest in *digital audio broadcasting* (DAB), a movement to change this has begun [1], [2]. DAB offers radio stations the ability to broadcast *compact disc* (CD) quality audio signals to their listeners while having the ability to offer a number of additional services. The HD Radio™ standard developed by iBiquity [3] is one of the TAB standards being used to deliver DAB. It uses *orthogonal frequency division multiplexing* (OFDM) signalling to transmit digital data using an *in-band on-channel* (IBOC) signalling scheme, thus allowing it to operate in conjunction with existing AM and FM transmissions.

OFDM is a multicarrier modulation that has attracted considerable attention from the communication industry in recent years. Its major advantage is that it is resistant to multipath fading effects [4]. The major drawback to OFDM is its high *peak-to-average power ratio* (PAPR). Its high PAPR makes it extremely sensitive to non-linear and memory distortions which may exist in the *high power amplifier* (HPA) stage of a transmitter [5].

The HD Radio™ standard permits both digital-only and hybrid analog-digital operating modes, thus allowing the co-existence of both existing analog stations and new digital stations in the same AM and FM bands. The necessity of the hybrid analog-digital signalling scheme is clearly evident when one looks at the situation from a consumer cost perspective, as all existing analog receivers would become obsolete if an abrupt change to an all digital broadcast standard occurred. In addition to the cost the consumer would face, there is a cost to the broadcaster. A system for broadcasting all-digital and hybrid analog-digital signals reusing as much of the existing analog only transmitter equipment is therefore of particular interest. As a result, designers of the new digital radio systems must make their designs compatible with transmitters that were designed for analog signal transmission. This results in the need to examine existing AM and FM analog hardware to determine where potential problems may exist in using digital signals with these systems.

This paper investigates the issues that arise in attempting to integrate new digital radio standards into existing analog radio transmitters. The specific case of iBiquity's HD Radio™ hybrid AM signalling is studied in detail. The impact of hybrid transmission in PDM-based amplifiers is examined by studying the spectrum regrowth effects. First, a signal processing model representing an *envelope elimination and restoration* (EER) amplifier is constructed. With this model, the parameters of an ideal EER amplifier that contribute to distortions in the digital radio signal are identified, and simulations of their effects on system performance are demonstrated. From these results, the most critical parameters that determine compliance with the AM spectral mask are identified. Specifically, it is established that in addition to delay mismatch between the envelope and phase paths, the cutoff frequency of the envelope reconstruction filter in the envelope path of the EER amplifier deserve special attention. When the ideal reconstruction filter is replaced with the realistic one, then specialized LMS-based compensation algorithms are deployed to reduce the spectrum regrowth at the output of the amplifier. The principle for this algorithm is that the cascade of the predistorter and the envelope reconstruction filter will have a better frequency response which will result in improved compliance with the spectrum mask of the overall system.

II. Problem Formulation

The majority of AM transmitters operating today are based on the EER amplification technique, originally proposed by Kahn in 1952 [6]. Because of its theoretical potential for high-efficiency operation (close to 100%), its choice as the basis for AM transmitters, which can have power levels up to 300kW, is apparent. Its implementation is relatively straightforward and is discussed below.

A. EER Architecture

Figure 1 shows a block diagram of a commonly used EER architecture. The basic approach in EER is to amplify the envelope signal separately from the phase processing and then recombine them afterwards to form an amplified version of the input signal. This could be done on a signal that has already been modulated with a *radio frequency* (RF) carrier or, as is the case here, a baseband signal. The RF carrier is added as part of the phase recovery process of this baseband signal.

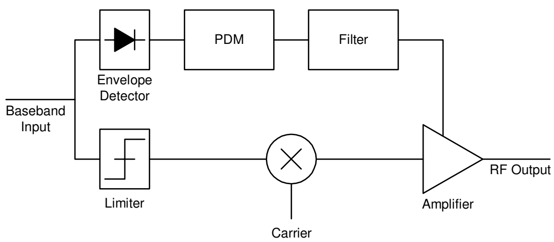

Fig. 1. EER System Block Diagram.

For the case shown in Fig. 1, the baseband input signal is split into two paths for processing. Considering the upper branch, the baseband signal passes through an envelope detector to remove the phase component of the signal. This envelope signal is then amplified and applied to the power supply of the output amplifier. The amplification is generally performed using *pulse duration modulation* (PDM) or some other highly power efficient means. The PDM pulses is used as the input to a switch (typically a *field-effect transistor* (FET)) that controls a high voltage source, thus creating a PDM replica with a much higher voltage level. To convert these pulses back into continuous voltage levels, a lowpass filter is applied to the signal in a similar fashion as a restoration filter is used in recovering a *pulse code modulation* (PCM) signal. Because the switch used has virtually no 'ON' resistance and almost infinite 'OFF' resistance, the amplification process is nearly 100% efficient.

In terms of the lower branch of the EER architecture in Fig. 1, the baseband signal is passed through a limiter to remove the envelope from the signal. This results in a constant envelope phase information signal. To add the RF carrier, this signal is mixed with a constant envelope phase signal that is derived from a sine wave signal at the desired carrier frequency. The resulting signal is then delayed to match any delay added to the envelope signal caused by the amplification process, mainly due to the filtering, and is then passed to the input of the output amplifier. As the input to the amplifier is of constant envelope, the problem of amplitude distortion caused by non-linear amplification does not exist.

The result of this combination of highly efficient envelope amplification and highly efficient phase recombining is a system with an overall power efficiency approaching 100%. This high efficiency is one of the EER amplification technique's major advantages. Combined with high linearity and relative simplicity, it was an obvious choice for use in AM transmitters. Unfortunately, EER amplifiers, as is the case with other types of amplifiers, face difficulties when dealing with signals with a high PAPR, and are unable to correct for non-linearities in their components [7]. This poses several challenges, such as dealing with spectral regrowth, when trying to incorporate digital signalling into systems employing EER.

It should be noted that the EER architecture presented in Fig. 1 is based on analog components. With the move to DAB, and digitizing the system, a number of the EER components could be replaced and their functions performed in digital hardware. The envelope detector, limiter, carrier addition and part of the PDM generation could all be performed digitally and converted to analog before being recombined in the amplifier. One of the issues involved in this process, the sampling frequency at which to operate the system, will be discussed in Section III.

B. Hybrid Digital-Analog AM Broadcast Standard

A brief description of the hybrid digital-analog AM signal is provided here to illustrate the changes that must be made to incorporate this new signalling scheme into existing AM transmitters. The signalling standard for the hybrid digital-analog AM system is outlined in [8], [9].

The hybrid AM signal has a bandwidth of 15kHz, which is much larger than the 5 or 8kHz bandwidth of the analog only AM signal. This is one potential issue when converting from the analog to the hybrid system, as existing amplifiers were designed for the narrow bandwidth of the analog AM signal and they may not handle the new hybrid AM signal.

To be compatible with existing AM receivers, the hybrid AM signal leaves the bandwidth occupied by the original analog signal (5 or 8kHz) unchanged, and adds the digital signalling in the remaining bandwidth. The digital signalling is broken down into three sidebands: primary, secondary and tertiary. Each sideband uses OFDM signalling, employing a different *quadrature amplitude modulation* (QAM) encoding on each subcarrier. This may result in signal with a high PAPR which could lead to spectral regrowth.

C. Approach

This paper will first examine the effects the envelope filter has on the hybrid AM signal by varying the filter's cutoff frequency and the delay matching between the two paths using MATLAB® simulations. As a consequence of these simulations, we will also discuss the effect of the sampling frequency used in the simulation as it has an impact on the simulation results and can be directly related to the issue of moving some of the EER components from the analog to digital domain (as mentioned earlier). Based on these results, digital signal processing is employed in the envelope path to enhance performance of the envelope reconstruction filter and to improve delay matching between the envelope and phase paths.

III. EER Parameter Effects

This section outlines the effects that the various parameters of the EER amplifier have on the hybrid AM signal used in DAB. The effects of the envelope filter bandwidth and the delay matching between the envelope and phase path are examined using MATLAB® simulations and synthetic hybrid AM data which was generated based on the hybrid AM specifications.

The MATLAB® model used to represent the EER system is illustrated in block format in Fig. 2. As can be seen from comparing this diagram to that of the general EER architecture in Fig. 1, the Envelope Detector and Limiter have been replaced with a Rectangular to Polar conversion block and the carrier addition has been removed. The model also works on a sampled version of the input signals as opposed to the continuous signals used in the actual EER system. A discrete-time system clock frequency is used to sample the continuous time analog signal $x(t)$ to represent it as a discrete time signal $x[n]$. The effects that result from the choice of this discrete-time system clock frequency will be elaborated on shortly.

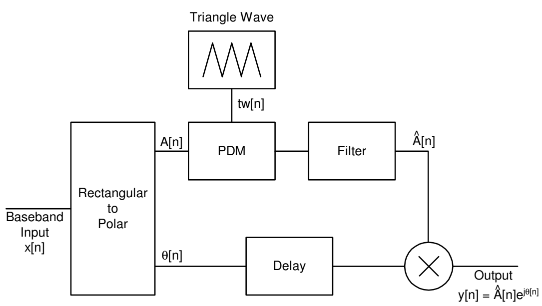

Fig. 2. Schematic of the MATLAB® Simulation of the EER System.

The PDM signal was generated using a triangular waveform with a repetition rate of 135kHz and the natural sampling method. Two PDM signals were generated, with one 180 degrees out of phase with the other, and the two were combined to form a biphase PDM signal. This is normally accomplished in an AM transmitter by having the output of several EER amplifier combined together with half of them using one PDM signal and the other half use the second PDM signal. This effectively doubled the repetition rate to 270kHz which moves the spectral images caused by the PDM process further away from the desired signal, making it easier to filter them out with the envelope filter. Because amplification of the PDM signal in simulation would have amounted to a multiplicative factor in the model, this step was not essential and therefore dropped from the model.

A. Sampling Frequency and Quantization Noise

With a model established, we used it to examine the effects that some of the EER parameters had on the hybrid AM signal. First we examined the effect of the choice of the discrete-time system clock frequency had on the system output. Referring back to Fig. 2, an ideal brickwall filter with a cutoff of 40kHz was used for the envelope filter and the delay block was set to perfectly match the delay introduced by the brickwall filter. Fig. 3 show the *power spectral density* (PSD) plots of the signal after it has been reconstructed at the output.

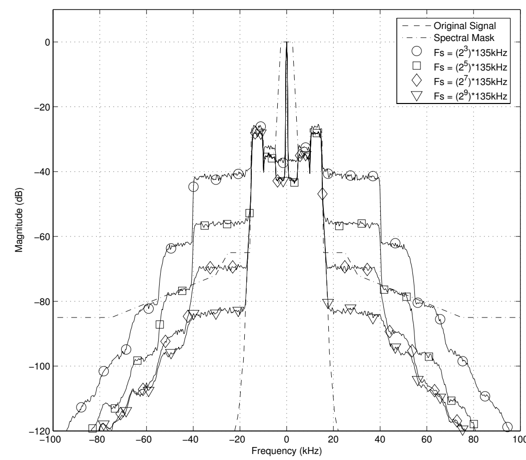

Fig. 3. PSD of Filtered PDM Signal with Phase Reconstruction for Various Sampling Frequencies.

The plot show the signals using discrete-time system clock frequencies of 1080kHz, 4320kHz, 17280kHz and 69120kHz or 2^3, 2^5, 2^7 and 2^9 times the PDM repetition rate respectively. The choice of these clock frequencies to be exactly power of 2 multiples of the PDM repetition rate was so that the results could be interpreted as being the result of the number of bits of resolution that existed in the PDM block. The number of bits of resolution translates into the total number of possible PDM pulse widths that can occur. This results in a quantization noise level that must be dealt with.

The effect of increasing the system sampling frequency can be clearly seen. For every two bits of resolution, the noise floor dropped by approximately 12dB, which was expected. The system clock determines the number of quantization levels representing the envelope values. In discretizing the signal levels there is 6db reduction of quantization noise for every additional bit in a sample's representation [10].

From this it can be concluded that the discrete-time system clock frequency must be at least 2^8 times that of the PDM repetition rate to observe any spectrum effects within the spectral mask and to lower the quantization noise floor sufficiently to be capable of meeting the spectral mask. As a result, a discrete-time system clock frequency of 2^8 times the PDM repetition rate was used in the rest of the simulations in this paper to capture spectral effects resulting from other imperfections in the simulated EER system. The choice of the discrete-time system clock frequency in the simulations to meet the frequency spectrum mask is also valid in determining the clock rate which would be necessary in the digital hardware implementation of the EER architecture mentioned in Section II. In digital generation of PDM, the pulse widths are determined in the hardware with a resolution equal to the clock resolution as opposed to comparing the analog signal to a triangle waveform resulting in an almost infinite resolution.

B. Filter Cutoff Frequency

The effect of the choice of the cutoff frequency of the filter used in the envelope path is examined in this subsection. Again, referring back to Fig. 2, the PDM stage was removed so that the quantization noise effects previously demonstrated would not be present. The filter was replaced with brickwall filters having different cutoff frequencies and the delay block was set to match each of these brickwall filter delays.

Figure 4 shows a PSD plot of the system output when the brickwall filter cutoff was varied from 40kHz to 100kHz, in steps of 20kHz. The overall system sampling frequency was set at 34560kHz. As can be seen from the plot, as the filter passband width was increased, the closer the system output signal followed the original input signal in the main portion of the passband. Increasing the filter passband width also lowered the spike that occurs at the transition of the passband to the stopband.

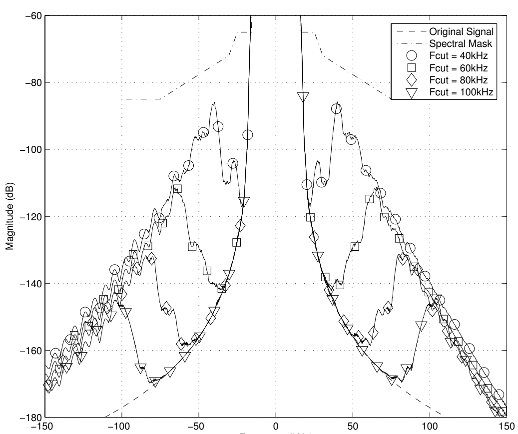

Fig. 4. PSD of the Output Signal when Various Filter Cutoffs Used.

It can be concluded from this that the wider the filter's passband, the better the performance of the system. However, there are some practical constraints which must be placed on this. The filter's passband should not exceed half the repetition rate in conventional PDM, and be no greater than the repetition rate in biphase PDM. In addition, the filter attenuation in the stopband should be high enough to suppress the spectral images of the PDM signal and the transition band should be as narrow as possible. As a result, for a filter of fixed order, there is a compromise between allowing sufficient bandwidth in the passband and large enough attenuation in the stopband.

C. Envelope and Phase Path Delay Mismatch

In this subsection, the effect of errors in the delay matching between the envelope and phase paths are demonstrated. As was used in the previous subsection, the PDM block in Fig. 2 was removed, but now the filter cutoff was fixed and the delay was adjusted.

Figure 5 shows a PSD plot of the system output when the discrete-time system clock frequency was 34560kHz, the brickwall filter cutoff was set at 40kHz, and the delay between the envelope and phase paths was varied ± 5 and ± 10 samples from the perfect match. As can be seen from the plot, as the delay varied from that of the perfect match, the spectral regrowth increases dramatically. For a delay mismatch of just 5 samples, the regrowth was approximately 15dB. Based on a discrete-time clock frequency of 34560kHz, a delay of 5 samples translates into a delay of approximately 145ns. It can be seen that only the magnitude of the delay has an impact on the spectral regrowth, with the direction of the delay being irrelevant. It is obvious from these results that the delay matching of the two data paths is critical to reducing the spectral regrowth of the system.

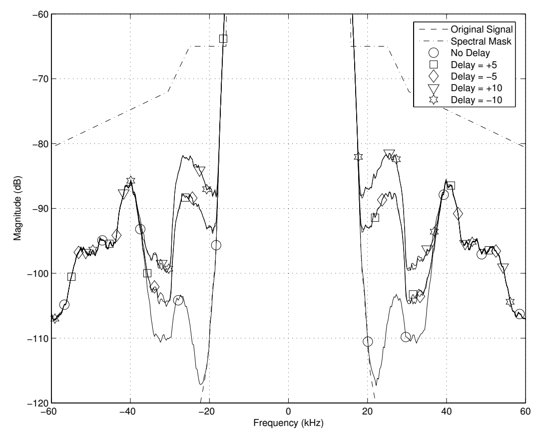

Fig. 5. PSD of the Output Signal with Various Delay Matching Errors.

D. Parameter Simulation Results

Based on the simulations and results presented above, the following conclusions were drawn:

- The discrete-time system clock frequency should be at least 2^8 times higher than the PDM repetition rate which translates to the PDM signal having at least 8 bits of resolution. This observation, determined by spectral mask fitting, is valid for both simulation and digital hardware implementations of PDM;
- The passband of the envelope reconstruction filter should be as wide as possible while maintaining good attenuation separation between the passband and stopband; and
- Delay matching between the envelope and phase data paths is critical.

These criteria were taken into account during the development of the linear compensation system discussed in Section IV.

IV. LMS Predistortion

In Section III, the simulation models used an ideal brickwall filter to remove the repeated spectral images that existed in the PDM signal. Obviously, a brickwall filter is not realizable in real hardware, especially since this is a high power, high voltage system, so an actual lowpass filter must be designed instead. In general, there are a number of factors that must be taken into account when designing a lowpass filter, and in this specific case, there are several additional constraints that must be met. As was outlined in Section III, the filter must have a wide passband so as to reduce spectral regrowth while still having sufficient attenuation in the stopband to remove the spectral images from the PDM signal. The desire for a linear phase response and the minimum amount of ripple in the magnitude response over the entire passband of the filter are also key. Linear phase response will make delay matching easier while a flat magnitude response in the passband will reduce distortion in the envelope signal.

In addition to these requirements, some additional issues must be handled due to the nature of the system. Because the filter exists after the PDM stage, which performs the amplification of the envelope signal, the filter must be able to handle the high voltage being output by the PDM. As such, the components used to design the filter must handle this, and as a result, the cost of these components can be high. In terms of a practical design for market, the minimization of cost is a goal. To that end, a filter consisting of the minimum number of components to meet the requirements of the system is desired.

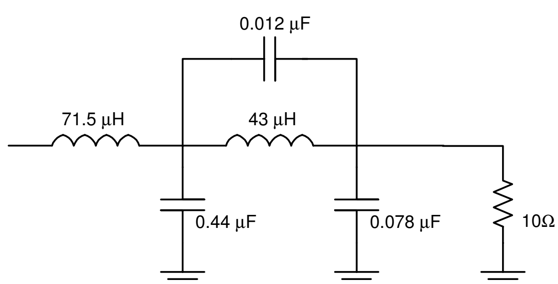

Fig. 6. Lowpass Filter Schematic.

Figure 6 shows the schematic for a lowpass filter that was designed based on a fourth order inverse-Chebyshev filter with its zero adjusted to fall at 135kHz to suppress the spectral image due to the PDM. The motivation for the choice of this filter was that it has a good magnitude response for a filter of such a low order and has a highly linear phase response in the passband [11]. The parameters of the filter were first designed based on the filter formula assuming a 10Ω load representing the the amplifier input impedance (which included effects of the antenna) and were later tuned on a real amplifier system to optimize the performance.

The filter has a passband of approximately 43kHz, based on the 3dB down point with approximately 1dB of ripple in the passband. It has a zero around 135kHz, thus drawing the magnitude response down significantly at that frequency. It also has approximately 25dB of attenuation between the passband and stopband. In terms of the phase response, it is relatively linear in the passband of the filter. Even thou this filter has fairly good characteristics, it still results in some undesirable spectral regrowth when the hybrid AM signal is passed through the model. To improve the filters response characteristics, an LMS based predistorter was added in front of the PDM stage so that the cascade of the predistorter and filter would have a better response.

A. Predistorter Design

In this paper, the predistorter was designed, using the LMS algorithm, 'off-line' as a FIR filter and once the coefficients were known, the fixed predistorter was added into the EER model. Random data was used, sampled at a frequency of 270kHz and the predistorter FIR filter length, L, was set at 31. The error updating factor, μ, was set at 0.015 and the number of training points, N, was set at 10000. These parameters were derived experimentally and the choice of these specific values was based on these results. Fig 7 shows a plot of the magnitude and phase responses for the lowpass filter shown in Fig. 6, the predistorter designed based on the LMS algorithm and the result after cascading the two together.

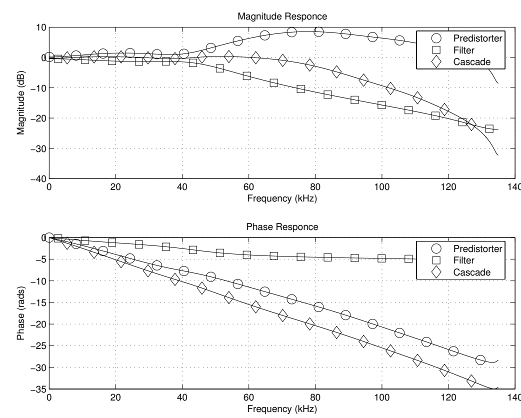

Fig. 7. Magnitude and Phase Responses of the Low Pass Filter, Predistorter and the Cascade of the two.

It should also be noted that the performance of LMS algorithm in terms of converging to the optimum solution for the compensation of the reconstruction filter may be degraded if the actual DAB signal was used in the training phase as that signal has a finite bandwidth where as the random data currently being used has a flat PSD. This may result in a need for a training signal to be used if this algorithm were to be done adaptively in the actual amplifier system as opposed to the 'off-line' method being used here. Because the predistorter coefficients were determined at a lower sampling rate that that used in the simulations, the filter coefficient of the predistorter were upsampled so that a representation of the filter at the sampling frequency of 34560kHz was generated.

B. Results of using the Predistorter

With the predistorter now designed, it was added into the simulation model shown in Fig. 2. It was placed before the

PDM generation stage where the signal was still in its low voltage state. The PSD results of the output signal with and without the predistorter are shown in Fig. 8. As can be seen, the predistorter improves the spectral regrowth of the overall system as noted by the absence of the lobe in the spectrum immediately adjacent to the original signal, i.e. the spectral mask requirement at 27kHz is now met by a 10dB margin as opposed to a 1dB margin in the system with no predistortion.

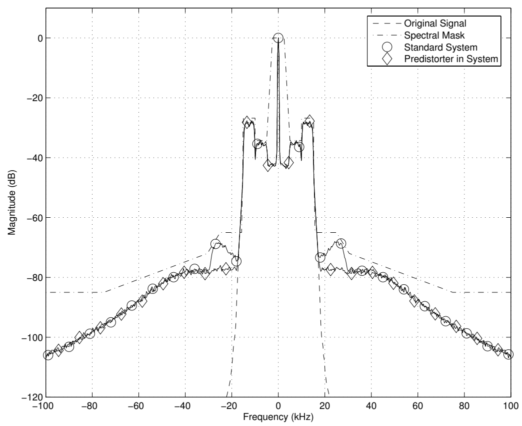

Fig. 8. PSD of the Output Signal Before and After the Predistorter was Added.

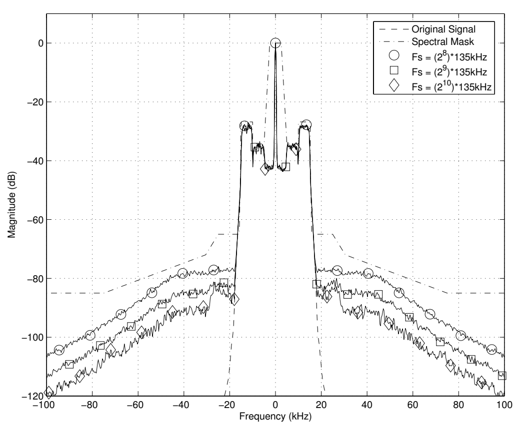

Fig. 9. PSD of the Output Signal with the Predistorter at Various Discrete-Time System Clock Frequencies.

Figure 9 shows the PSD results from simulations similar to those used to obtain the PSD plots in Fig. 8, by varying the discrete-time system clock. Specifically, the discrete-time system clock frequency was set at 34560kHz, 69120kHz and 138240kHz which correspond to 2^8, 2^9 and 2^{10} time the PDM repartition rate respectively. This can also be looked at as the system using 8, 9 and 10 bits of resolution in the PDM sample representation and results in reduced quantization effects (floor) that potentially could mask the benefits of using the predistorter. Looking at the 27kHz point in Fig. 9 it can be seen that with a discrete-time system clock frequency of 34560kHz, a 10dB margin exists between the signal and the spectrum mask. This margin increases to 15dB when the discrete-time system clock frequency is increased to 69120kHz and 19dB when the discrete-time system clock frequency is set at 138240kHz. This demonstrates the high potential of the predistorter to improve the spectral performance of the EER amplifier. It also shows the existence of the trade off between spectral efficiency and the system clock rate if one pursues digital implementation of PDM in the EER amplifiers.

C. LMS Predistortion Results

This section has outlined the design of a lowpass filter to replace the idealized brickwall filter used in the previous simulations and has demonstrated the effects this filter has on the spectral regrowth of the system output. It has also proposed the use of a predistorter designed using the LMS algorithm to reduce the spectral regrowth effects caused by this real filter and has shown the results of using this predistorter in the system in conjunction with the real filter.

V. CONCLUSIONS

This paper examined the issues of implementing the HD Radio™ standard developed by iBiquity on existing AM broadcast equipment based on the EER amplification technique. First the various components of the EER amplifier were examined to identify the most critical components in terms of their effect on the spectral regrowth of the hybrid digital-analog AM signal. From this it was identified that the parameters of the envelope filter were the most critical in determining the system performance.

Based on this analysis, and LMS-based predistorter was proposed to improve these filter parameters. Simulations were performed to verify the performance of this solution and proved to be highly effective in reducing the spectral regrowth in the system.

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